

CASE STUDY — DATA ANALYSIS

Stock monitoring & optimization for a network of health centers

Cleaning of heterogeneous data, building of management indicators (consumption, rotation, stockouts) and operational recommendations for management — from a raw monthly record of 5 centers.

January – March 2026
Period analyzed

5 centers
18 monthly records

Excel > Analysis > PDF
Methodology

July 2026
Report by Mahoussi



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01 — FRAMING

Context & objectives

This project was carried out as a portfolio case study for MAHOUSSEI DATA INSIGHTS, to demonstrate a complete data-handling methodology — from the raw file to recommendations — applied to a common challenge for health-center networks: monitoring stock availability.

A network of 5 health centers submits a monthly record of its product stock (opening stock, closing stock, number of stockouts observed). Without consolidation, these monthly records — entered independently by each center — remain hard to compare and do not allow supply tensions to be anticipated before they become stockouts.

Source data: a raw table of 18 monthly records (5 centers x 3 months, with 3 additional records), with 5 fields: center, month, opening stock, closing stock, stockouts.

Objectives

- **Clean the raw data** — fix input inconsistencies (center names, date formats, missing values) before any analysis.
- **Consolidate the 5 centers over 3 months** — obtain one comparable view, whereas each record was entered separately.
- **Build management indicators** — consumption, stock rotation rate, stockout rate, by center and by month.
- **Produce a dashboard and recommendations** — directly usable by management to prioritize its actions.

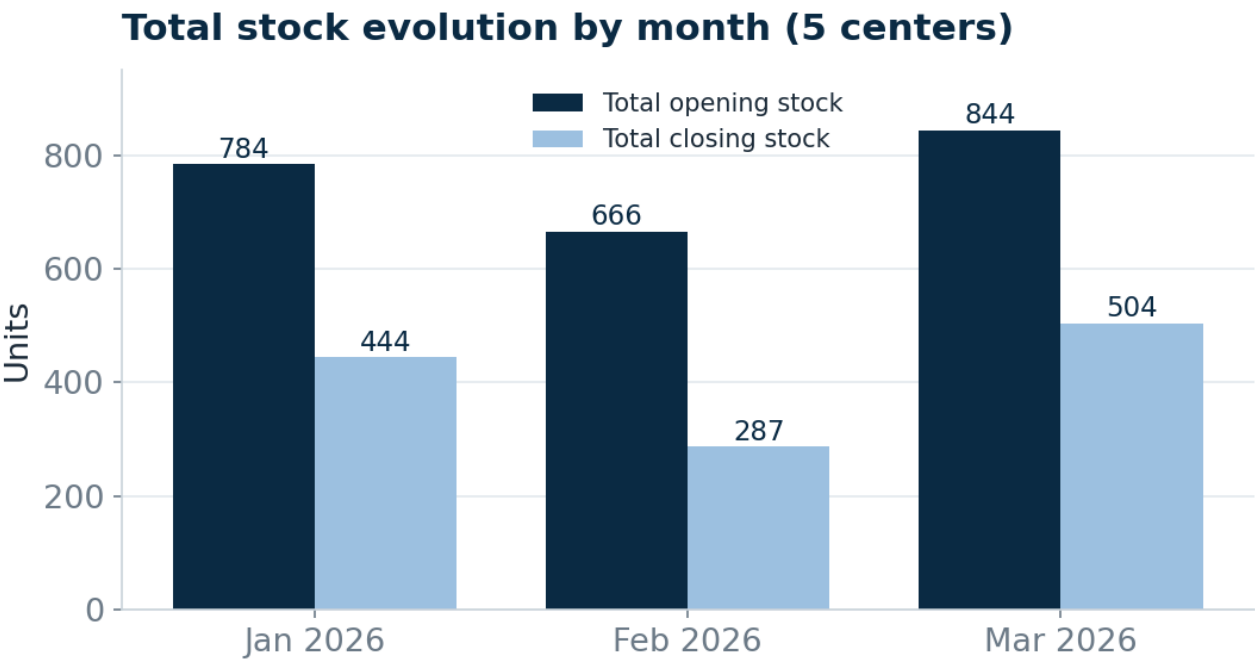
Approach: Excel (cleaning & formulas) > consolidated analysis > dashboard > recommendations.

02 — OVERVIEW

Executive summary

Over the 3 months analyzed, the network remains broadly supplied, but one clear warning signal stands out on a single center — and consumption is accelerating.

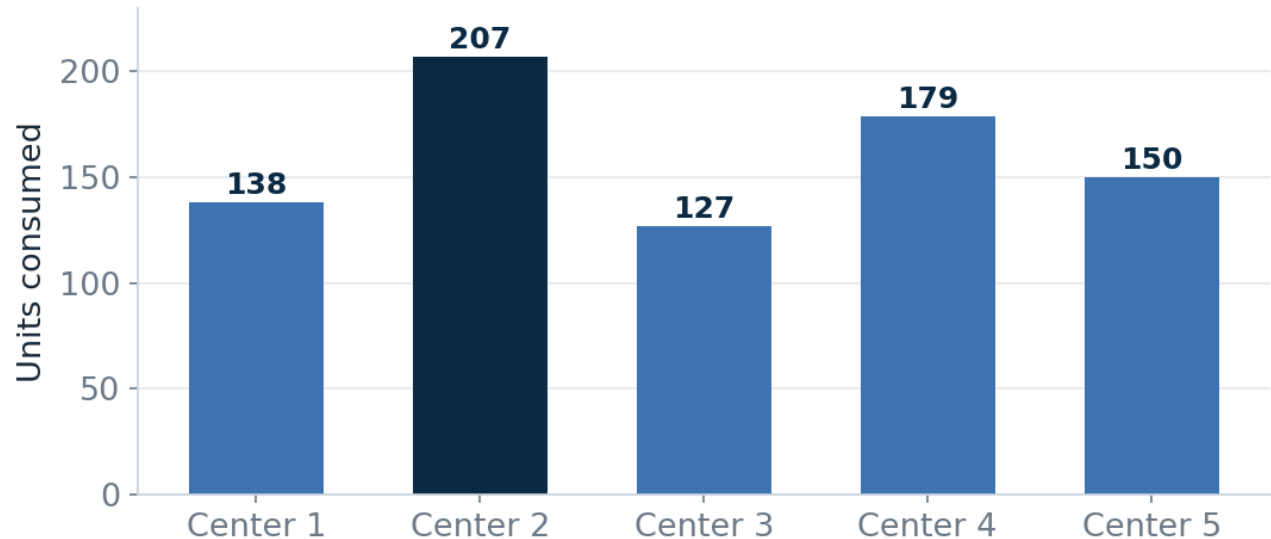
MOST CRITICAL CENTER Center 5 Only center with a stockout (25% rate)	STOCK REMAINING (END M ARCH) 1,235 units available across the network	TOTAL CONSUMPTION 801 units over 3 months (Jan-Mar)	AVERAGE STOCKOUT RATE 5.6% 1 stockout in 18 records
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Closing stock stays clearly positive every month — no center ended a month at zero, apart from Center 5's one-off episode.

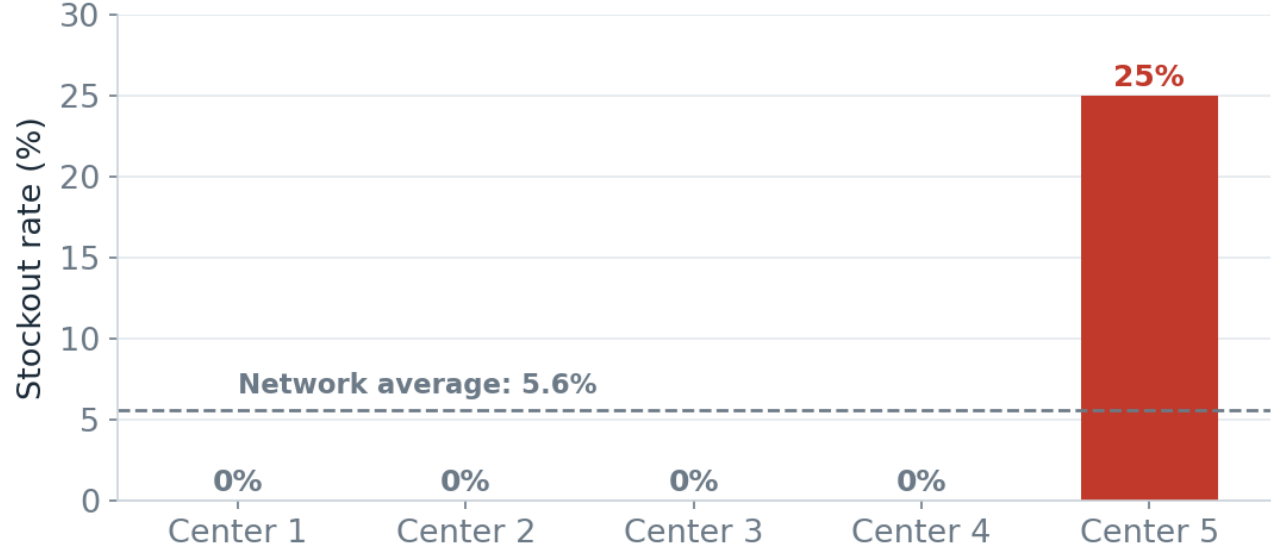


Total consumption by center (3 months)



Center 2 accounts for 26% of the network's total consumption.

Stockout rate by center



Center 5 is the only one to deviate from the network average (5.6%).



03 — ANALYSIS

Detailed results

Consolidated view by center then by month, from the 18 cleaned records.

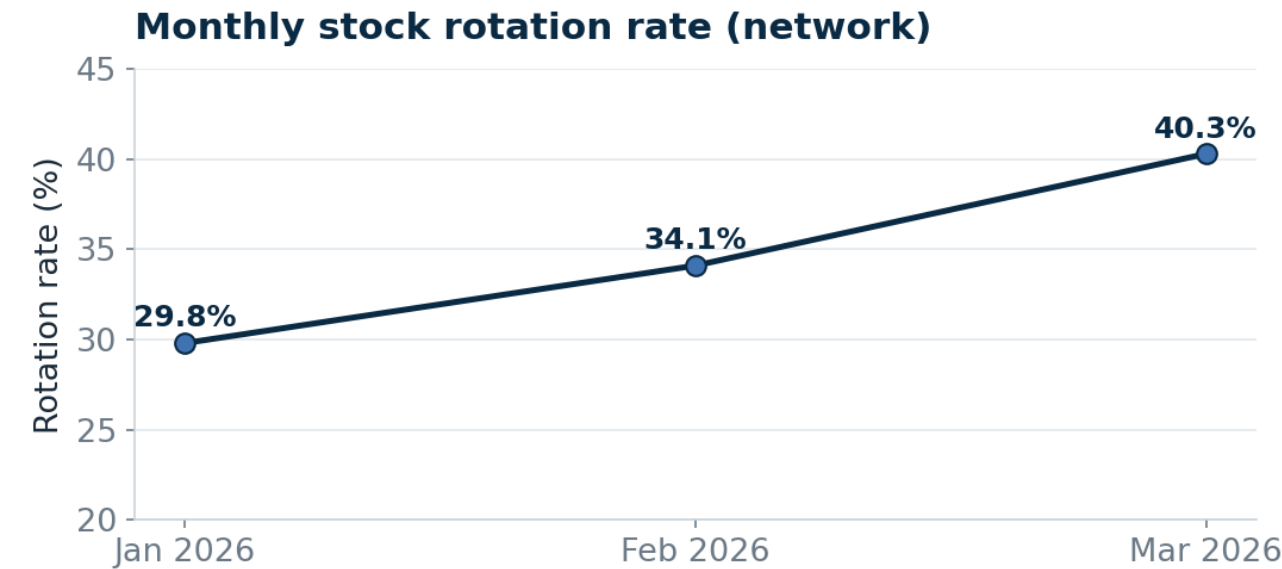
Summary by center

Center	Opening	Closing	Consumption	Records	Stockouts	Rate
Center 1	332	194	138	3	0	0%
Center 2	523	210	207	4	0	0%
Center 3	615	291	127	4	0	0%
Center 4	516	337	179	3	0	0%
Center 5	308	203	150	4	1	25%
Network total	2,294	1,235	801	18	1	5.6%

Summary by month

Month	Opening	Closing	Consumption	Stockouts	Consumption change
January 2026	784	444	234	0	—
February 2026	666	287	227	1	-3.0%
March 2026	844	504	340	0	+49.8%

Stock rotation



Monthly rotation rate (consumption / opening stock) — the network rises from 29.8% in January to 40.3% in March, i.e. consumption growing 45.3% faster than available stock over the period.

Rotation rate by center (total consumption / total opening stock)

Center 1	Center 2	Center 3	Center 4	Center 5	Network
41.6%	39.6%	20.7%	34.7%	48.7%	34.9%

04 — TAKING ACTION

Strategic recommendations

5 levers prioritized from the analysis results, framed for operational management.

1. Secure Center 5's supply

Finding: Across the 18 records, Center 5 accounts for the trimester's only stockout episode, with a 25% stockout rate (1 record in 4) versus 0% for the other four centers.

Recommended action: Move this center to weekly monitoring instead of monthly, and trigger a restock as soon as an alert threshold — to be defined with the logistics team — is crossed.

2. Anticipate Center 2's needs, the network's top consumer

Finding: Center 2 used 207 units over 3 months, i.e. 26% of total network consumption, for a 39.6% rotation rate of its opening stock.

Recommended action: Prioritize it in the supply schedule to prevent it from tipping into a tension zone if its consumption keeps rising.

3. Watch the upward consumption trend

Finding: Total network consumption rises from 234 to 340 units between January and March 2026 (+45.3%), while available opening stock grows only 7.7% over the same period.

Recommended action: Check, over a longer history, whether this rise reflects seasonal demand or a structural trend, and adjust future orders before a lasting gap forms between consumption and supply.

4. Improve data collection at the source

Finding: 3 stock values (2 closing, 1 opening) were missing in the raw records, and both center names and dates were entered in heterogeneous formats (up to 4 variants for one center).

Recommended action: Standardize data entry (drop-down list for centers, a single date format) and enforce a mandatory check before validating each monthly record.

5. Generalize monthly rotation tracking

Finding: The rotation rate ranges from 20.7% (Center 3) to 48.7% (Center 5) across centers — a gap revealing very different stock-management situations.

Recommended action: Add this rate as a standard monthly dashboard indicator, with a per-center alert threshold, to spot overstock or tension early — before a stockout.

05 — UNDER THE HOOD

Methodology & data quality

Before any analysis, the 18 raw records were audited and cleaned. Below are the inconsistencies identified and how they were handled.

4 center-name formats

e.g. "CENTRE-1", "Centre 2", "centre2", "centre_5" — unified into "Center 1" ... "Center 5".

3 different date formats

Free text ("january", "Jan 2026", "february...") and Excel serial numbers (46023, 46054, 46082) — standardized to "mmm yyyy".

3 missing values

2 closing-stock and 1 opening-stock values missing — kept as absent data rather than arbitrarily estimated, so as not to distort the indicators.

Once the fields were standardized, two derived indicators were computed for each complete record:
Consumption = opening – closing · **Consumption rate** = consumption ÷ opening · **Stockout rate** = stockouts ÷ records.

Records with a missing stock value were excluded from the corresponding consumption calculation, but kept in the original totals — to stay faithful to the data actually available rather than filling gaps with assumptions.

06 — APPENDIX

Before / after cleaning

The first 8 records, as received from the client, compared line by line with their cleaned and enriched version.

BEFORE — raw data received from the client

CENTER	MONTH	START	END
CENTRE-1	january	108	85
Centre 2	46023	106	N/A
CENTRE-3	46023	135	86
Centre 4	Janv 2026	219	146
CENTRE-5	Janv 2026	81	41
CENTRE-1	46054	119	72
centre2	46054	104	39
Centre 3	Fev 2026	197	N/A

AFTER — data cleaned by the Data Analyst

CENTER	MONTH	START	END	CONS.	RATE
Center 1	Jan	108	85	23	21.3%
Center 2	Jan	106	—	—	—
Center 3	Jan	135	86	49	36.3%
Center 4	Jan	219	146	73	33.3%
Center 5	Jan	81	41	40	49.4%
Center 1	Feb	119	72	47	39.5%
Center 2	Feb	104	39	65	62.5%
Center 3	Feb	197	—	—	—

The remaining 10 records (March 2026 and additional records) were processed the same way and are included in the totals in section 03.

07 — IN SUMMARY

Conclusion

This case shows, on a deliberately imperfect dataset, a reproducible working method: audit and clean the data before any calculation, build simple but actionable indicators (consumption, rotation, stockouts), then translate the results into concrete decisions rather than plain figures. The same approach applies to multi-store sales, warehouses, or any periodic multi-site tracking.

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